Adnan Harun Doğan

Personal Information

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Employment

Feb 2023–Present — **Research Assistant**, Image Processing and Pattern Recognition Laboratory (ImageLab), METU
Supervisor: Prof. Dr. Sinan Kalkan, Assoc. Prof. Dr. Emre Akbas — *Ankara, Turkey*

Jan 2025–Oct 2025 — **Intern**, Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich
Supervisor: Prof. Dr. Christian Holz — *Zürich, Switzerland*

Oct 2021–Feb 2022 — **Intern**, Center for Artificial Intelligence and Robotics (ROMER), METU
Supervisor: Assoc. Prof. Dr. Erol Sahin, Assoc. Prof. Dr. Bugra Koku — *Ankara, Turkey*

Jun 2021–Sep 2021 — **Intern**, Cyber-Physical Systems Laboratory (AICPS), University of California, Irvine
Supervisor: Prof. Dr. Mohammad Abdullah Al-Faruque — *Irvine, California, USA*

Education

2022–2025 — **Master of Science in Computer Engineering**, Middle East Technical University (METU)
Advisor: Prof. Dr. Sinan Kalkan, Co-advisor: Assoc. Prof. Dr. Emre Akbas — *Ankara, Turkey*
Thesis: Incorporating Piecewise Linear Functions with Constant Regions in Backpropagation

2018–2022 — **Bachelor of Science in Computer Engineering**, Middle East Technical University (METU)
Senior Project: ReQuest: Automated Question-Answer Generation from Text — *Ankara, Turkey*

Selected Graduate Coursework

Optimization: Linear Optimization (IE553), Numerical Optimization (IAM566), Optimization Methods for ML (IAM771)
Probability and Statistics: Probability Theory (IAM541), Decision Theory and Bayesian Analysis (STAT565)
Deep Learning: Deep Learning (CENG501), Advanced Deep Learning (CENG502), Deep Generative Models (CENG796)

Research Interests

My research has focused on gradient-based learning through non-differentiable operations. I am interested in problems where the model or its loss contains a discrete decision (e.g., ordering detections by confidence, assigning predictions to targets, choosing which layers a token passes through, or deciding where to refine a mesh). Such decisions give backpropagation little or no gradient signal. I have designed structure-respecting surrogate gradients for the building blocks behind them, such as ranking, sorting, matching and routing, and studied how this choice shapes what the network learns.

Keywords: gradients through discrete decisions, conditional computation, neural state estimation, wearable biosignal modeling, differentiable ranking and matching, physiological event detection

Publications

Peer-reviewed publications

10. **Dogan, A. H.**, Demirel, B. U., & Holz, C. (2026). Frequency-weighted neural Kalman filters. ***International Conference on Robotics and Automation (IEEE ICRA)***. arXiv:2606.02251 (PDF, Code)
9. **Dogan, A. H.**, Deniz, M., Alemdar, H., & Baran, O. U. (2026). CoFINN: Conservation flux informed neural networks for physics problems governed by conservation laws. *Scientific Reports*. arXiv:2607.06587
8. Demirel, B. U., **Dogan, A. H.**, Rossie, J., Möbus, M., & Holz, C. (2025). Beyond subjectivity: Continuous cybersickness detection using EEG-based multitaper spectrum estimation. ***IEEE Transactions on Visualization and Computer Graphics (IEEE TVCG), 31***(5), 2525-2534. doi.org/10.1109/TVCG.2025.3549132 (arXiv, Code)
7. Yavuz, F., Cam, B. C., **Dogan, A. H.**, Oksuz, K., Kalkan, S., & Akbas, E. (2024). Bucketed ranking-based losses for efficient training of object detectors. ***European Conference on Computer Vision (ECCV)***, Lecture Notes in Computer Science, vol 15117, pp. 93-109. doi.org/10.1007/978-3-031-73202-7_6 (arXiv, PDF, Code)
6. Amin, M. A., **Dogan, A. H.**, Kuru, E. S., Sever, Y., & Angin, P. (2024). Misuse detection and response for orchestrated microservices based software. *International Conference on Advanced Information Networking and Applications (AINA)*, Lecture Notes on Data Engineering and Communications Technologies, vol 204, pp. 217-226. doi.org/10.1007/978-3-031-57942-4_22 (Data)
5. Karagoz, P., Cekinel, R. F., **Dogan, A. H.**, Oktay, B., Ozturk, A. U., Tonay, T. S., & Tuncel, B. M. (2023). Enhancing underground built heritage analysis with text mining: A case study on Cappadocia. In M. Akkar Ercan & K. Aydiner (Eds.), *Valorising Underground Built Heritage in Cappadocia* (pp. 61-92). CNR Edizioni. ISBN 978-88-8080-605-9. (PDF)
4. Sever, Y., & **Dogan, A. H.** (2023). A Kubernetes dataset for misuse detection. *ITU Journal on Future and Evolving Technologies, 4*(2), 383-388. doi.org/10.52953/fplr8631 (PDF, Code)
3. Sever, Y., Ekinci, G., **Dogan, A. H.**, Alparslan, B., Gurbuz, A. S., Jabrayilov, V., & Angin, P. (2022). An empirical analysis of IDS approaches in container security. *International Workshop on Secure and Reliable Microservices and Containers (SRMC)*, pp. 18-26. doi.org/10.1109/SRMC57347.2022.00007 Best Paper Award
2. Demirel, B. U., **Dogan, A. H.**, & Al-Faruque, M. A. (2021). Two might do: A beat-by-beat classification of cardiac abnormalities using deep learning and domain-specific features. *Computing in Cardiology (CinC)*, pp. 1-4. doi.org/10.23919/cinc53138.2021.9662935 (PDF, Code)
1. Buyukbas, E. B., **Dogan, A. H.**, Ozturk, A. U., & Karagoz, P. (2021). Explainability in irony detection. *Big Data Analytics and Knowledge Discovery (DaWaK)*, Lecture Notes in Computer Science, vol 12925, pp. 152-157. doi.org/10.1007/978-3-030-86534-4_14

Manuscripts under review

2. **Dogan, A. H.**, Arasli, D., Oksuz, K., Akbas, E., & Kalkan, S. (2027). *Variational surrogate Jacobians: A framework for constrained gradient propagation.* Under Review[†].
1. **Dogan, A. H.**, Celtik, S. O., Baran, O. U., & Alemdar, H. (2026). *Prediction-driven adaptive mesh refinement with graph neural networks for transonic airfoil flows.* Under Review at Journal of Aerospace Science and Technology.

Theses

1. **Dogan, A. H.** (2025). *Incorporating piecewise linear functions with constant regions in backpropagation.* M.Sc. Thesis, Middle East Technical University. (Thesis)

[†] Venue withheld while under double-blind review.

Undergraduate Research Projects

Sep 2021–Jun 2022

High Performance Intrusion Detection and Prevention for Software-Defined Container Networks in the Cloud, METU

CONTSEC (TUBITAK 3501): security of containerized microservices on Kubernetes

- Designed and implemented an intrusion detection pipeline for containerized cloud systems
- Built a public Kubernetes misuse dataset of network flows from 10 labelled attack scenarios

Advisor: Assoc. Prof. Dr. Pelin Angin *Ankara, Turkey*

Feb 2021–Jun 2022

Image Processing and Pattern Recognition Laboratory (ImageLab), METU

Conservation flux informed neural networks and adaptive mesh refinement for transonic airfoils

- Designed an HLLC Riemann solver momentum loss on pixels, cutting drag error by 15%
- Trained CNN, FNO, ViT and diffusion models with and without the momentum flux loss
- Refined GNN airfoil meshes at Ducros-flagged shocks, cutting the area-based RMSE by 12%

Advisor: Assist. Prof. Dr. Hande Alemdar, Assist. Prof. Dr. Ozgur Ugras Baran *Ankara, Turkey*

Oct 2020–Jun 2021

Textual Event Graph HUB (TEGHub), METU

Textual analysis for irony detection and research impact prediction

- Adapted T5 to Turkish irony detection, its first use for the task, reaching 59.5% accuracy
- Fine-tuned BERT with Simple Transformers, reaching 69.0% accuracy on Turkish irony
- Crawled scientific papers and their Twitter mentions as data for research impact prediction
- Applied LDA, K-Means and rule-based VADER sentiment to 1,742 Cappadocia reviews

Advisor: Prof. Dr. Pinar Karagoz *Ankara, Turkey*

Feb 2020–Jun 2021

School of Medicine, Koç University

Healthcare policy simulation for evidence-based decision making

- Used simulation models focusing on mechatronics applications in the healthcare domain
- Built the citation graph of 999 health systems papers from Microsoft Academic Graph data
- Developed GCN-based node embeddings on this citation graph of health systems literature

Advisor: Assist. Prof. Dr. Ozge Karanfil *Istanbul, Turkey*

Teaching

Teaching Assistant, Computer Engineering Department, METU Feb 2023–Present

8. CENG 280 - Automata Theory (Spring 2024, Spring 2025, Spring 2026)
7. CENG 242 - Programming Language Concepts (Spring 2023, Spring 2025, Spring 2026)
6. CENG 223 - Discrete Computational Structures (Fall 2023, Fall 2024, Fall 2025)
5. CENG 424 - Logic for Computer Science (Fall 2023, Fall 2025, Fall 2026)
4. CENG 460 - Introduction to Robotics for Computer Science (Fall 2024)
3. CENG 213 - Data Structures (Spring 2024)
2. CENG 382 - Analysis of Dynamical Systems (Spring 2023)
1. CENG 334 - Introduction to Operating Systems (Spring 2023)

Undergraduate Teaching Assistant (Volunteer), Computer Engineering Department, METU Spring 2019–Fall 2021

3. CENG 240 - Programming with Python for Engineers (Fall 2021)
2. CENG 111 - Introduction to Computer Engineering Concepts (Fall 2020)
1. CENG 230 - Introduction to C Programming (Spring 2019)

Honors and Scholarships

2022–2023	TUBITAK 2247-C STAR'23 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2022	Graduated in the first quantile of the class by Middle East Technical University (METU)
2021–2022	TUBITAK 2247-C STAR'22 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2020–2021	TUBITAK 2247-C STAR'21 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2017	University Entrance Exam: ranked 1466th of 2.27M candidates by OSYM, Turkey

Computer Skills

Languages:	Python, C/C++, Rust
Tools & Libraries:	PyTorch, CVXPYLayers, Gurobi, OR-Tools, POT, MNE, ROS2
DevOps & HPC:	GitHub, Docker, Slurm (Workload Manager), Apptainer

Languages

Turkish:	Native
English:	Fluent (TOEFL iBT 109/120, 2022)